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This project consists of the investigation of a series of fragile glass formers, including phosphates, chalcogenides, and aluminates, with the aim to elucidate structural developments associated with glass formation. To this end we use a combination of concurrent Raman and Brillouin light scattering as the structural probe, and molecular dynamics simulations to interpret the experimental observations. Raman scattering allows us to monitor changes in the symmetry of structural building blocks, while Brillouin scattering yields the complex mechanical modulus at the molecular scale. The storage component of this modulus provides a measure of the degree of connectivity between building blocks and the structural integrity, whereas the loss modulus assesses the energy dissipated in aperiodic motions of these building blocks. The role of simulation is to explain the observed measures in terms of structural changes that occur during glass formation.

	This methodology constitutes a new approach towards investigating glass structure and dynamics. In particular, it provides insights into the elusive intermediate-range order of non-crystalline structures. Variable parameters in this study include the nature of the network former and the chemical composition of the glass, and the preparation history, i.e., sol-gel synthesis vs. melt processing.
Task Description:	1. Devise and refine a methodology, based on concurrent Brillouin and Raman scattering, to characterize structural developments in fluids, sol-gel derived systems, and supercooled melts during the glass formation process. While we have already demonstrated that the scattering techniques can act as a structural probe on the basis of comparative evaluation, we expect that the combination with molecular dynamics simulations will allow for a direct mechanistic and structural interpretation of experimental observables. 2. Substantiate and elucidate the concept of polyamorphism, i.e., the existence of more than one thermodynamically and structurally distinct amorphous state for a given material. Specifically, we will verify whether polyamorphic transitions are ubiquitous to glass formation. 3. By understanding the fundamental principles that underlie glass formation and that control structural developments in glass-forming systems, develop knowledge that allows for a more targeted design of amorphous materials with specific properties. With this project we take a novel approach towards investigating the amorphous state of matter, and therefore expect to gain unprecedented insights in this subject matter. Accordingly, we expect to (i) advance fundamental aspects of glass science, (ii) devise knowledge for the creation of new glass-forming materials with applications in optical waveguides, photonic crystals, optical amplifiers, lasers. sensors, integrated circuits, and logic devices, and (iii) develop a technique of investigation that is of general interest for NASA's microgravity research program because it allows for the contact-free measurement of visco-elastic properties of fluids, i.e., without obliterating the environment free of external forces provided by microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While ultimately container-less melting in microgravity conditions provides the ideal environment for investigating structural developments in deeply supercooled glass-forming melts or in a sol-gel precursor maintained in delicate thermo-mechanical equilibrium, the method of investigation developed in this research can be applied in numerous situations on earth in which direct contact between specimen and the mechanical probe is difficult or impossible. Such situations include specimens that are (i) chemically aggressive, (ii) held at very high temperature, (iii) of fragile constitution (e.g., gels), (iv) of small dimensions, and (v) part of a living organism. The ability to create glasses with lower optical losses or with an increased capacity to host lasing ions will advance the photonics and optical telecommunication industries. Furthermore, the non-invasive measurement of elastic properties and viscosity coefficients at the nano-scale has application in areas ranging from materials processing to bio-medical diagnostics.
	The work involved four aspects: 1. Instrumentation, experimental setup, and data analysis procedure 2. Monitoring structural developments of amorphous porous network structures during sol-gel synthesis using concurrent Brillouin (BLS) and Raman (RLS) light scattering 3. Monitoring structural developments of supercooled glass-forming melts using concurrent Brillouin (BLS) and Raman (RLS) light scattering 4. Molecular dynamics (MD) simulations of structural transitions in glasses and their crystalline counterparts. Progress regarding instrumentation, experimental, and analytical capabilities: A six-pass tandem Fabry-Perot interferometer (JSR Scientific instruments), which is used for BLS measurements and triple monochromator Raman spectrometer, TR777 (Princeton Instruments – Acton Research) were acquired and set up in our laboratory on either end of an optical table, with sample stages (one open stage for ambient temperature and one enclosed by a furnace for high temperature measurements), so that light scattered from the focal volume of the incident laser beam can be collimated and routed to either spectrometer, or when using a beam splitter, to both spectrometers simultaneously. We developed data analysis software. This software is specifically designed for the interpretation of the Brillouin spectra in terms of the generalized hydrodynamic theory, i.e., to extract both elastic moduli and viscosity coefficients from this data. The analysis procedure involves deconvolution to rid the spectra of instrument broadening and the fitting of a theoretical line shape to the deconvoluted data. The software is designed to run on unix platforms in an X-window environment. Finally, we developed a new high-temperature sample cell. Porous networks via sol-gel synthesis: We focused our experimental investigation on phosphate glasses, specifically, a series of glass compositions in the SiO2-P2O5 system. Phosphosilicate glasses have recently received significant attention for potential application as the electrolyte material for proton exchange membranes. These glasses exhibit the highest proton conductivity of any all-oxide material. This is due to the superb protogenic properties of phosponic groups and to the fact that these glasses form a unique structure, in which the silica and phosphorous pentoxide components are phase separated at the nano-scale. The two phases arrange into two interpenetrating networks; the silica-rich one provides mechanical stability and the phosphate-rich one provides percolating paths for proton transport. To clearly identify the role of network percolation in ionic conductivity, it was essential that we could produce monolithic gels and xerogels, and study the mechanical properties and ionic conductivities. Hence, significant effort was spent on preparing monolithic gels in the form of thin disks. Using concurrent BLS and RLS, we have carried out a systematic study of the gelation process and monitored structural developments of the sol-gel derived materials upon drying, between gelation point and fully dense xerogel. Curiously, and contrary to our expectations, the gelation point is not revealed by either the elastic storage or loss modulus.

Task Progress:	The principal objective of this investigation was to establish the relationship between pore structure, pore dynamics, and ionic conductivity. To this end, we compare the information obtained form BLS and DIS. Both the modulus and the conductivity increase monotonously with P2O5 concentration. The rate of increase is large at low concentrations but decelerates significantly upon reaching the 50% mark. The increase of conductivity with P2O5 concentration is to be expected, because the proton donating function is being attributed to this compound. The fact that the elastic modulus increases commensurately with the conductivity is a bit more surprising in that high stiffness is typically associated with rigid and immobile structures. To accentuate this observation, we compare the proton conductivity with the reciprocal of the mechanical loss tangent, i.e., M'/M", where M' is the storage and M" the loss modulus. Accordingly, the conductivity increases for as long as the elastic storage modulus increases more rapidly than the loss modulus. This indicates that the structural and chemical changes responsible for the enhanced stiffness of the network also cause a relative degrease in the energy dissipation experienced by mobile structural moieties, e.g., protons or proton donating groups. In other words, with increasing P2O5 content, the pore structure in the xerogel adjusts so as to reduce frictional forces on small migrating species. Hence, the increase in proton conductivity is not only due to the increase in concentration of proton donating groups, but also an increase in proton mobility, at least at low P2O5 concentrations. Glass formation in supercooled melts: Brillouin light scattering allows one to determine the high-frequency visco-elastic properties of glass-forming liquids as a function of temperature. Unlike conventional methods, the Brillouin technique requires no mechanical contact with the specimen, and can be used for measurement under extreme conditions. The frequency shift between Brillouin and Rayleigh peaks, omega, and the velocity of sound, c0. Based on the theory of linear visco-elasticity, one can relate the frequency shift between Brillouin and Rayleigh peak to the elastic storage modulus and the width of the Brillouin peak to the viscosity or loss modulus. While the storage component is a measure of structural integrity (e.g., degree of networking in a silicate melt), the loss modulus reflects the amount of energy lost due to anelastic deformations involving structural imperfections, and gauges their mobility. In the course of this project we investigated a series of alkali germanate and alkali phosphate glass-forming liquids, measuring Brillouin spectra as a function of temperature and composition. Generally, we found that (i) higher alkali concentrations result in stiffer glass network, due to the reinforcing effect of the strongly ionic alkali cations. It also result in a higher loss modulus and viscosity, because of the reduced mobility and compliance of the network segments. In the series of K, Na, and Li, the reinforcing effect increases because of the increasing field strength associated with the decreasing ionic radii. The activation energies and pre-exponential coefficients characterizing the structural relaxation mechanism also increase with increasing alkali concentration. However, for a given alkali to phosphate ratio, these quantities are quite comparable regardless of the nature of the alkali cation. Molecular simulations: Using molecular dynamics simulations based on the reactive force field we develop in our group we pursued to objectives: one is to gain a better understanding of the structural characteristics of these porous xerogel networks by generating structural models using simulations of the formation process, and two is to provide insights into the structural dynamics of these materials based on studying the mechanical and kinetic responses of the models so generated. In the course of the research, we also improved our reactive force field, specifically in its ability to simulate systems that exhibit variable coordination (e.g., 4- or 6-coordinated Si and 3- or 4-coordinated B). The most relevant results pertain to (i) insights concerning structural details of gel networks that developed under different chemical conditions and (ii) the mechanisms of network deformation and how this can enhance ion mobility. Three distinct growth regimes for silica gels emerge, depending on the solvent-to-silica ratio: compact nano-particles, percolated silica networks, and branched clusters. These growth regimes can be identified based on distinctive structural features. In the case of compact particles the radial distribution functions exhibit a broad maximum that coincides with the radius of gyration of the aggregates, while in continuous networks the radial distribution function increases steadily beyond the near-range structural features. Furthermore, these growth regimes can be distinguished based on the concentrations of structural defects, such as dangling bonds and residual OH groups. Key to network structural rearrangements, which always occur without breaking or forming of chemical bonds, and the ensuing property changes, are local rearrangements that are facilitated by relatively small pivotal rotations about Si-O or B-O bonds. These rotations, which occur abruptly and can be described as a type of buckling deformation, are not only responsible for the anomalous thermo-mechanical behaviors described above (i.e., network stiffening upon heating and softening upon compression), but they are also responsible for of the negative thermal expansion in some silica and silicate structures. The significance of these transitions for ionic transport in network structures and the design of solid electrolyte materials is due to the fact that providing a mechanism for the network structure to easily contract creates space for the mobile charge carrying species to migrate at a low activation energy.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
Articles in Peer-reviewed Journals	Huang L, Durandurdu M, Kieffer J. "New B2O3 crystals predicted from concurrent molecular dynamics simulations and first-principles calculations." Journal of Physical Chemistry C 2007 Sep;111(37):13712-20. http://dx.doi.org/10.1021/jp0735583 , Sep-2007
Articles in Peer-reviewed Journals	Bhattacharya S, Kieffer J. "Molecular dynamics simulation study of growth regimes during polycondensation of silicic acid: from silica nano-particles to porous gels" Journal of Physical Chemistry C 2008 Feb;112(6):1764-71. http://dx.doi.org/10.1021/jp073808f , Feb-2008